



Morfeusz Sphere: Geometric Energy Storage at the Edge of the Universe

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Abstract

This paper presents the hypothesis of the Morfeusz Sphere - a geometric structure at the edge of the Universe, composed of quasi-antimatter (qAn). Within the ontology based on quantized spacetime (p-gluons), we postulate the existence of a thin boundary layer with thickness 155 000 ly and energy density $5 \times 10^{-5} \text{ J m}^{-3}$. Verification calculations show that the quasi-annihilation process in this layer generates power of $4.1 \times 10^{51} \text{ W}$, representing over 100-fold excess compared to the energy requirements of the cosmic microwave background (CMB). The model offers a physical mechanism explaining the origin of dark energy and the energy increase in the expanding Universe, providing an alternative to the standard Λ CDM model.

1 Introduction

Within the geometric ontology of spacetime, we postulate the existence of quasi-antimatter (qAn) as a state of spacetime with negative geometric polarization. According to the conservation principle of baryon number for geometric states:

$$N_{qAn} + N_{An} = N_{qM} + N_M = \text{const}$$

assuming $N_{An} \approx 0$, all missing antimatter manifests as qAn. In this model, qAn has been pushed to the periphery of the Universe in the process of Great Segregation, creating a thin boundary layer that serves as the source of dark energy.

Philosophical Depth of the Model

Although the Morfeusz Sphere model originated as a purely geometric mathematical construction, its consequences lead to a profound philosophical intuition. The structure in which the Universe possesses:

- **A stable boundary** (Morfeusz Sphere) separating it from the exterior
- **Internal metabolism** (quasi-annihilation) providing energy
- **Homeostasis** (constant CMB temperature despite expansion)

remarkably resembles the structure of living organisms.

This analogy was not the starting point but an emergent consequence of the model. It suggests that **life may be a more fundamental property of reality** than commonly believed, manifesting at all levels of existence - from the cell to the cosmos.

2 Initial Parameters

2.1 Radius of the Observable Universe

$$R_{\text{oW}} = 4.4 \times 10^{26} \text{ m} \quad (\approx 46.5 \text{ Gly})$$

2.2 Energy Densities

- Baryonic matter: $\rho_M = 8.0 \times 10^{-11} \text{ J m}^{-3}$
- Dark matter (quasi-matter): $\rho_{qM} = 4.2 \times 10^{-10} \text{ J m}^{-3}$
- **Total:** $\rho_{M+qM} = 5.0 \times 10^{-10} \text{ J m}^{-3}$
- Quasi-antimatter (in the layer): $\rho_{qAn} = 5.0 \times 10^{-5} \text{ J m}^{-3}$

3 Calculations

3.1 Volume of the Observable Universe

$$V_{\text{ow}} = \frac{4}{3}\pi R_{\text{ow}}^3 = \frac{4}{3}\pi(4.4 \times 10^{26} \text{ m})^3 = 3.57 \times 10^{80} \text{ m}^3$$

3.2 Total Matter Energy

$$E_{M+qM} = \rho_{M+qM} \cdot V_{\text{ow}} = (5.0 \times 10^{-10} \text{ J m}^{-3})(3.57 \times 10^{80} \text{ m}^3) = 1.785 \times 10^{71} \text{ J}$$

From conservation principle: $E_{qAn} = E_{M+qM} = 1.785 \times 10^{71} \text{ J}$

3.3 Volume Occupied by qAn

$$V_{qAn} = \frac{E_{qAn}}{\rho_{qAn}} = \frac{1.785 \times 10^{71} \text{ J}}{5.0 \times 10^{-5} \text{ J m}^{-3}} = 3.57 \times 10^{75} \text{ m}^3$$

3.4 Thickness of the qAn Layer

We assume that qAn forms a spherical shell surrounding the observable Universe:

$$V_{qAn} = \frac{4}{3}\pi[(R_{\text{ow}} + \Delta R)^3 - R_{\text{ow}}^3]$$

Substituting numerical values:

$$3.57 \times 10^{75} \text{ m}^3 = \frac{4}{3}\pi[(4.4 \times 10^{26} \text{ m} + \Delta R)^3 - (4.4 \times 10^{26} \text{ m})^3]$$

Calculating the volume of the observable Universe:

$$\frac{4}{3}\pi R_{\text{ow}}^3 = \frac{4}{3}\pi(4.4 \times 10^{26} \text{ m})^3 = 8.5184 \times 10^{79} \text{ m}^3$$

Solving the equation:

$$(4.4 \times 10^{26} \text{ m} + \Delta R)^3 = 8.5184 \times 10^{79} \text{ m}^3 + 3.57 \times 10^{75} \text{ m}^3$$

$$(4.4 \times 10^{26} \text{ m} + \Delta R)^3 = 8.5184 \times 10^{79} \text{ m}^3(1 + 1.0 \times 10^{-5})$$

Taking the cube root of both sides:

$$4.4 \times 10^{26} \text{ m} + \Delta R = 4.4 \times 10^{26} \text{ m} \cdot (1 + 1.0 \times 10^{-5})^{1/3}$$

Using the approximation $(1 + x)^{1/3} \approx 1 + \frac{x}{3}$ for $x \ll 1$:

$$4.4 \times 10^{26} \text{ m} + \Delta R \approx 4.4 \times 10^{26} \text{ m} \cdot (1 + 3.333 \times 10^{-6})$$

Solving for ΔR :

$$\Delta R \approx 4.4 \times 10^{26} \text{ m} \cdot 3.333 \times 10^{-6} = 1.467 \times 10^{21} \text{ m}$$

4 Result

$\Delta R = 1.47 \times 10^{21} \text{ m} \approx 155\,000 \text{ ly}$

5 Visualization: Morfeusz Sphere

To comprehend the proportions of this extremely thin boundary layer, let us perform a comparative scaling.

Imagine that the entire observable Universe along with its surrounding Morfeusz Sphere has been reduced to the size of a sphere with Earth's diameter (12 742 km). In this scale, the "shell" of quasi-antimatter, responsible for accelerating cosmic expansion, would have a thickness of only:

$$\text{Scale} = \frac{1.2742 \times 10^7 \text{ m}}{8.8 \times 10^{26} \text{ m}} = 1.448 \times 10^{-20}$$

$$\Delta R_{\text{scale}} = (1.47 \times 10^{21} \text{ m}) \times (1.448 \times 10^{-20}) = 0.213 \text{ m}$$

21.3 cm

This is approximately:

- the height of a large cooking pot
- the thickness of a yoga mat
- the length of a forearm

Analogous to a **Dyson Sphere** surrounding a star, this hypothetical boundary layer of qAn can be called the **Morfeusz Sphere** - constituting an ultra-thin boundary between the known Universe and the geometric "void" of quasi-antimatter.

6 Discussion

The calculated thickness of the qAn layer is approximately **155 thousand light-years**, which corresponds to:

- the typical thickness of a spiral galaxy disk
- the size of a large dwarf galaxy

On a cosmological scale, this is an **extremely thin boundary layer**, justifying the use of the term "shell" rather than "halo" or "foam". Even after drastic scaling (where the entire Universe becomes the size of Earth), the qAn layer remains extremely thin compared to the object it surrounds.

The density value $\rho_{qAn} = 5 \times 10^{-5} \text{ J m}^{-3}$, being 5 orders of magnitude greater than the average dark energy density in the Λ CDM model, corresponds to an intermediate state between basic spacetime (RKP) and fully formed antimatter (ZKP), confirming the concept of qAn as "failed antimatter" trapped in an intermediate state of geometric polarization.

7 Energy Verification of the Model - DeepSeek Calculations

7.1 Observational Data - CMB

7.1.1 Basic Parameters

$$\begin{aligned} T_0 &= 2.7255 \text{ K} \quad (\text{CMB temperature}) \\ k_B &= 1.380\,649 \times 10^{-23} \text{ J K}^{-1} \\ \hbar &= 1.054\,571\,8 \times 10^{-34} \text{ J s} \\ c &= 299\,792\,458 \text{ m s}^{-1} \\ t_{\text{universe}} &= 13.787 \times 10^9 = 4.354 \times 10^{17} \text{ s} \\ R_{\text{oW}} &= 4.398 \times 10^{26} \text{ m} \end{aligned}$$

7.1.2 CMB Density and Energy

$$\begin{aligned} \rho_{\text{CMB}} &= \frac{\pi^2}{15} \cdot \frac{k_B^4 T_0^4}{\hbar^3 c^3} \approx 4.645 \times 10^{-14} \text{ J m}^{-3} \\ E_{\text{CMB,total}} &= \rho_{\text{CMB}} \cdot V_{\text{oW}} \approx 1.656 \times 10^{67} \text{ J} \end{aligned}$$

7.2 Power Balance

7.2.1 Power Required to Maintain CMB

$$P_{\text{avg}} = \frac{E_{\text{CMB,total}}}{t_{\text{universe}}} \approx 3.803 \times 10^{49} \text{ W}$$

7.2.2 Power Generated by Quasi-Annihilation

$$\begin{aligned}
A_{\text{boundary}} &= 4\pi R_{\text{ow}}^2 \approx 2.431 \times 10^{54} \text{ m}^2 \\
d &= 0.01 \cdot \Delta R = 1.467 \times 10^{19} \text{ m} \\
V_{\text{active}} &= A_{\text{boundary}} \cdot d \approx 3.566 \times 10^{73} \text{ m}^3 \\
E_{\text{active}} &= \rho_{qAn} \cdot V_{\text{active}} = 5.0 \times 10^{-5} \text{ J m}^{-3} \cdot 3.566 \times 10^{73} \text{ m}^3 = 1.783 \times 10^{69} \text{ J} \\
P_{\text{QA}} &= \frac{E_{\text{active}}}{t_{\text{universe}}} \approx 4.095 \times 10^{51} \text{ W}
\end{aligned}$$

7.2.3 Losses Due to Universe Expansion

$$\begin{aligned}
H_0 &= 2.184 \times 10^{-18} \text{ s}^{-1} \quad (\text{Hubble constant}) \\
P_{\text{loss}} &= H_0 \cdot E_{\text{CMB, total}} \approx 3.617 \times 10^{49} \text{ W}
\end{aligned}$$

7.3 Power Comparison

$$\begin{aligned}
\frac{P_{\text{QA}}}{P_{\text{avg}}} &= \frac{4.095 \times 10^{51}}{3.803 \times 10^{49}} \approx 107.7 \\
\frac{P_{\text{QA}}}{P_{\text{loss}}} &= \frac{4.095 \times 10^{51}}{3.617 \times 10^{49}} \approx 113.2
\end{aligned}$$

7.4 Process Efficiency

$$\eta = \frac{P_{\text{loss}}}{P_{\text{QA}}} = \frac{3.617 \times 10^{49}}{4.095 \times 10^{51}} \approx 0.0088 = 0.88\%$$

8 Conclusions

- Power generated by quasi-annihilation ($4.095 \times 10^{51} \text{ W}$) is **108 times greater** than the power needed to maintain CMB
- The model is **energetically consistent** - sufficient power reserve exists
- The quasi-annihilation process requires efficiency at the level of **0.88%** to power CMB
- **99.12%** of generated power can drive the accelerated expansion of the Universe
- The Morfeusz Sphere with thickness 155 000 ly constitutes a stable energy source

Summary

Calculations confirm the feasibility of the Morfeusz Sphere model as an energy source for the Universe's background radiation. The energy balance is positive, and the geometry of the sphere allows for long-term stability of the quasi-annihilation process. The model offers a physical mechanism explaining the origin of energy in the expanding Universe, which is missing in the standard Λ CDM model.

8.1 Implications for Universe Evolution

- **Constant CMB temperature:** The model predicts maintenance of $T_0 = 2.725$ K despite expansion
- **Powering expansion:** The remaining $\approx 99.12\%$ of quasi-annihilation power can drive accelerated expansion
- **Long-term stability:** The energy reserve in the Morfeusz Sphere ensures mechanism operation for time greatly exceeding the Universe's age

8.2 Advantages Over the Standard Model

- **Solution to the energy problem:** Provides a physical mechanism for energy increase in the expanding Universe
- **Active vs Passive:** CMB as a stationary state rather than a relic
- **Falsifiability:** Generates specific predictions regarding temperature evolution and structure correlations

References

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